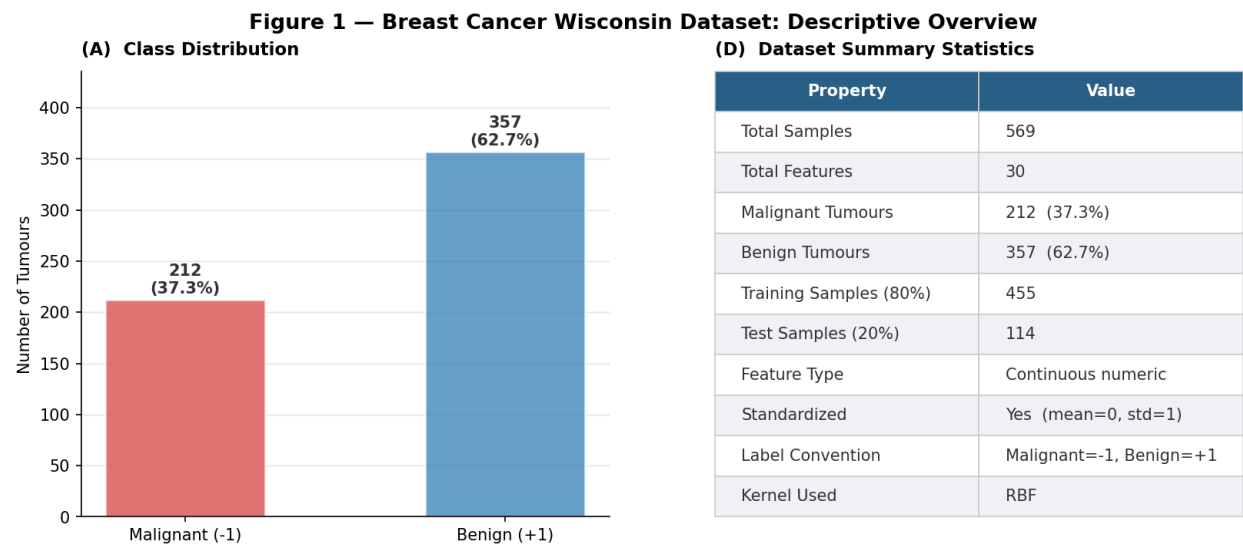


Research Question: Can a support vector machine accurately classify breast tumors as malignant or benign, and how does the choice of C affect its performance?

Describe the Problem and Dataset:

One of the key difficulties in applying support vector machines (SVMs) to medical classification tasks is the choice of the regularization parameter C, which balances the bias-variance tradeoff. A large value of C makes SVM minimize misclassification on the training data by reducing the margin, resulting in higher model complexity and the risk of overfitting. A small value of C makes SVM maintain a wider margin and allows for some misclassifications, which may cause underfitting [4]. While overfit models can give accurate predictions on the data, their predictive power on new tumor samples will be low; conversely, underfit models may not even recognize malignant tumors. The goal of this project is to perform a systematic analysis of how the bias-variance trade-off controlled by C affects the classification accuracy of SVMs on the dataset, which would provide the optimal regularization parameter to improve the generalizability of SVM to new tumor samples. While SVM has presumably been used for the classification of breast cancer cases, this project aims to provide a detailed analysis of the bias-variance trade-off problem in the context of SVM application to medical data classification.

The dataset that was used in this project was the Breast Cancer Wisconsin dataset, created by Dr. William H. Wolberg at the University of Wisconsin-Madison and provided by the UCI Machine Learning Repository. Each sample comprises 30 features computed based on digitized images of fine-needle aspirates of breast masses, describing characteristics such as the radius, texture, and smoothness of the cell nuclei. The dataset contains 569 samples, of which 212 are malignant, and 357 are benign [6]. The data were divided into 80% for training and 20% for testing. Before training, all features were standardized so that their means equal zero and their standard deviations equal one. This prevents any single feature from dominating the margin calculation.



Machine Learning Algorithm Used to Analyze the Dataset

SVM is suitable for analyzing this dataset since it is a binary classification where the task is to classify tumors between benign and malignant. With only 569 samples, the dataset is relatively small. Generally, SVMs work best on small to medium datasets, unlike neural networks, which typically require large amounts of data to train reliably. The SVM model also works well with high-dimensional features. In this case, we have 30 features in total.

Some assumptions of SVM include linear separability. The algorithm usually performs best when a clear margin separates the training points. When the boundary is nonlinear, a kernel function can be applied to map the data into a higher-dimensional space where linear separation is possible. Feature scaling is also another key assumption for this model. This assumes that features are normalized or standardized, since larger features are more likely to dominate the margin calculation [1]. The model assumes no heavy class imbalance [2]. In this dataset, there are 212 malignant tumors and 357 benign tumors, indicating a mild imbalance. The model also assumes that the features are relevant and clean, and that there is a sufficient margin between the classes.

SVM is a supervised learning algorithm that finds the optimal hyperplane maximizing the margin between the two classes, in this case, malignant and benign. SVM seeks the optimal hyperplane defined by the equation $w \cdot x + b = 0$. Once the hyperplane is defined, we want to maximize the margin between the hyperplane and the nearest data points from each class. One way to achieve this is through minimizing $\frac{1}{2}\|w\|^2$. But we do not account for all the training points when determining the optimal unique boundary. Only the support vectors will be considered. Support vectors are training points closest to the decision boundary; they are the only points that can influence the boundary. To find the orientation and position of the unique boundary, we need to calculate w and b [3].

The Primal:

$$\text{Minimize: } \frac{1}{2}\|w\|^2$$

$$\text{Subject to: } y_i(w^T x_i + b) \geq 1 \quad \text{for all } i$$

However, solving for w and b can be difficult when there are more features than training samples. Considering that the value of w depends on the number of features in the dataset, it can be computationally expensive. In this dataset, with 30 features and 455 training samples, solving for w requires optimizing in a 30-dimensional space. The Lagrangian function helps overcome this problem by converting the primal to the dual, simplifying the calculation process. The Lagrangian is a combined function that incorporates each constraint directly into the objective by multiplying it by a new variable, α_i . The constraints that used to be on the side can now be simplified into a single mathematical equation [3].

The Lagrangian Function:

$$L(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_i \alpha_i [y_i(w^T x_i + b) - 1]$$

Once we have the Lagrangian function, we can differentiate with respect to w and b to get the KKT conditions. The KKT conditions serve as a checklist of five rules that must all be satisfied simultaneously at the optimal solution, and provide the stationarity conditions used to eliminate w and b from the Lagrangian [5].

KKT Conditions:

Stationarity w :

$$\partial L / \partial w = 0 \rightarrow w = \sum_i \alpha_i y_i x_i$$

Stationarity b :

$$\partial L / \partial b = 0 \rightarrow \sum_i \alpha_i y_i = 0$$

Using those expressions for w and b , we then substitute them back into the Lagrangian. The dual solves for the alpha values for each training point, and from there, we can determine our support vectors. The dual is more efficient because most alphas will be zero, resulting in only a few support vectors. Second, in the dual form, the optimization requires n alpha values, but not w for each feature, and it makes the method more efficient than the primal when the number of features is larger than the number of training points. Third, in the dual form, only inner products are needed. The RBF kernel is the default for most SVM models since it projects the data into an infinite-dimensional space when the shape is unknown [7]. In this dataset, with 30 features and 569 training samples, the dual formulation is preferred as it solves for 569 alpha values rather than optimizing in a 30-dimensional feature space [3].

The Dual Function:

$$W(\alpha) = \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j (x_i^T x_j)$$

Results from the algorithm

The SVM algorithm was trained and evaluated across six different values of C (0.001, 0.01, 0.1, 1.0, 10.0, 100.0) using an RBF kernel. Performance was measured using accuracy, AUC, recall, and precision. At small values of C (0.001, 0.01), both produced an accuracy of 0.6316 despite achieving a recall of 1.000. At $C = 0.1$ and $C = 1.0$, the accuracy increased to 0.9474 and 0.9825. At a C value of 10.0, the accuracy score continued to increase, reaching 0.9737. At C value 100.0, the accuracy score decreased back to 0.9474.

The best overall result was obtained with $C = 10.0$, yielding an AUC of 0.9957, an accuracy of 0.9737, a precision of 0.9859, and a recall of 0.9722. At the same time, there were 84 support vectors, indicating that the margin is not too wide and that only a few borderline cases remain within it.

As for values higher than $C = 10.0$, there appears to be no improvement, rather there seems to be a small decrease in performance of the model. Specifically, the accuracy dropped to 0.9474, while the number of support vectors stabilized at 64. This might indicate overfitting, as the margin is now too narrow to generalize to other tumors.

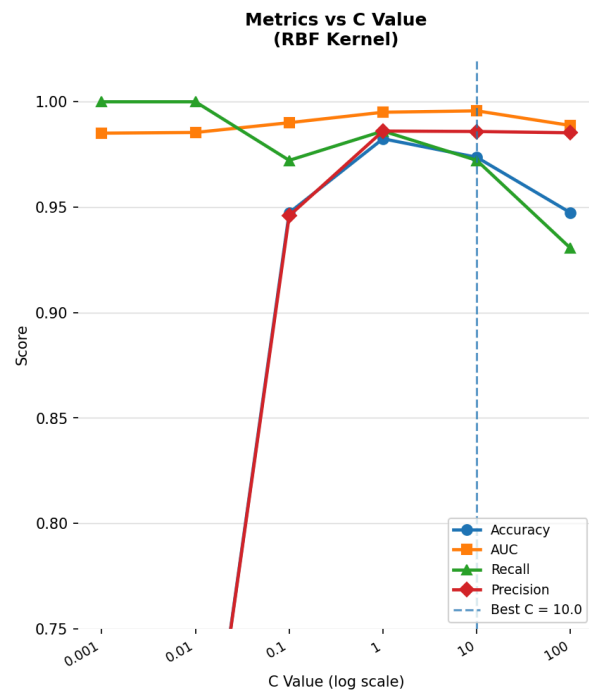
Five-fold cross-validation confirmed these findings. $C = 10.0$ achieved the highest mean AUC of 0.9945 (± 0.0072), followed by $C = 1.0$ at 0.9938 (± 0.0084). At very small values of C , the mean AUC dropped to 0.9841 and 0.9842 for $C = 0.001$ and $C = 0.01$, respectively. At $C = 100.0$, mean AUC declined to 0.9900 (± 0.0104), suggesting increased variance in model performance and an indication of overfitting.

Smaller C values yield larger margins that generalize well, while larger C values yield smaller margins that fit the training dataset better; however, there may be overfitting or underfitting with small and large C values, respectively. The best value for C for this dataset is 10.0, which yielded an excellent AUC score compared to others.

SVM Results — Effect of C on Classification Performance

C	Accuracy	Precision	Recall	AUC	Support Vectors
0.001	0.6316	0.6316	1.0000	0.9851	340
0.01	0.6316	0.6316	1.0000	0.9854	341
0.1	0.9474	0.9459	0.9722	0.9901	199
1.0	0.9825	0.9861	0.9861	0.9950	97
10.0	0.9737	0.9859	0.9722	0.9957	84
100.0	0.9474	0.9853	0.9306	0.9888	64

★ Best $C = 10.0$ | AUC = 0.9957 (highlighted)



References

- [1] Bhagyashrilakhadive. (2023, November 23). *Introduction to support vector machine(SVM)*. Medium.
<https://medium.com/@bhagyashrilakhadive/introduction-to-support-vector-machine-svm-4946083b8052>
- [2] Brownlee, J. (2020, January 30). *Cost-Sensitive SVM for Imbalanced Classification*. Machine Learning Mastery.
<https://machinelearningmastery.com/cost-sensitive-svm-for-imbalanced-classification/>
- [3] MLMath.io. (2019, February 16). *Math behind Support Vector Machine(SVM)*. Medium.
<https://ankitnitjsr13.medium.com/math-behind-support-vector-machine-svm-5e7376d0ee4d>
- [4] Sampaio, C. (2023, April 21). *Understanding SVM Hyperparameters*. Stack Abuse.
<https://stackabuse.com/understanding-svm-hyperparameters/>
- [5] Sasu, D. (2021, May 27). *Support Vector Machines — Lecture series — Karush-Kuhn-Tucker conditions part 1*. Medium.
<https://davidsasu.medium.com/support-vector-machines-lecture-series-karush-kuhn-tucker-conditions-part-1-259f4e096add>
- [6] UCI Machine Learning. (2016). *Breast Cancer Wisconsin (Diagnostic) Data Set*.
Www.kaggle.com. <https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data>
- [7] Visually Explained. (2022, May 9). *The Kernel Trick in Support Vector Machine (SVM)*.
Www.youtube.com. <https://www.youtube.com/watch?v=Q7vT0--5VII>